Supplementary Material for "Fixed Heavy-Tailed Probabilistic Forecasting under Missing Historical Observations"

[AUTHOR INPUT REQUIRED]

1 Full Fixed-Mask Matrix

The following tables retain all fixed-mask deterministic evaluations by dataset. Only historical inputs are masked; targets are unchanged.

Table 1: Full ETTh1 fixed-mask matrix.

Mask	Rate	MSE	MAE	RMSE
Block	0.10	0.388818	0.416915	0.623552
Block	0.30	0.423837	0.451188	0.651027
Block	0.50	0.486504	0.501242	0.697498
Block	0.70	0.591697	0.571655	0.769219
Channel	0.10	0.479410	0.455063	0.692394
Channel	0.30	0.583704	0.511084	0.764006
Channel	0.50	0.798862	0.627325	0.893791
Channel	0.70	0.900947	0.683061	0.949182
Mixed	0.10	0.486209	0.464128	0.697287
Mixed	0.30	0.495055	0.473908	0.703601
Mixed	0.50	0.553485	0.525226	0.743966
Mixed	0.70	0.715806	0.617339	0.846053
Random	0.10	0.378811	0.408069	0.615476
Random	0.30	0.425394	0.450336	0.652223
Random	0.50	0.525856	0.523115	0.725160
Random	0.70	0.688520	0.617690	0.829771

Table 2: Full Weather fixed-mask matrix.

Mask	Rate	MSE	MAE	RMSE
Block	0.10	0.158302	0.218956	0.397872
Block	0.30	0.180420	0.245178	0.424759
Block	0.50	0.210474	0.277453	0.458774
Block	0.70	0.254572	0.322297	0.504551
Channel	0.10	0.191289	0.238851	0.437366
Channel	0.30	0.284471	0.316355	0.533358
Channel	0.50	0.377902	0.393866	0.614738
Channel	0.70	0.495709	0.490618	0.704066
Mixed	0.10	0.175815	0.230739	0.419303
Mixed	0.30	0.224939	0.281431	0.474278
Mixed	0.50	0.306930	0.357264	0.554012
Mixed	0.70	0.400193	0.440502	0.632608
Random	0.10	0.169631	0.232543	0.411863
Random	0.30	0.240810	0.309629	0.490724
Random	0.50	0.331313	0.391138	0.575598
Random	0.70	0.434543	0.474367	0.659199

2 TimePerceiver Seed-Level Summaries

The four probability-head seeds are 2021, 2022, 2023, and 2025 on fixed official backbone checkpoints. The panels summarize the dataset-level means and variability; the main manuscript preserves all paired NLL deltas.

3 Distribution-Head Ablation Details

The ablation retains Gaussian, Laplace, fixed-df Student-t, and learned-df Student-t results. The learned variant is visible even though it is not the final method.

Table 3: Full Electricity fixed-mask matrix.

Mask	Rate	MSE	MAE	RMSE
Block	0.10	0.143247	0.243795	0.378480
Block	0.30	0.184630	0.289155	0.429686
Block	0.50	0.270835	0.369677	0.520418
Block	0.70	0.464169	0.520260	0.681300
Channel	0.10	0.213972	0.281108	0.462571
Channel	0.30	0.391038	0.403166	0.625331
Channel	0.50	0.569116	0.525639	0.754398
Channel	0.70	0.748916	0.649662	0.865399
Mixed	0.10	0.165759	0.254616	0.407135
Mixed	0.30	0.246785	0.325159	0.496775
Mixed	0.50	0.375182	0.437690	0.612521
Mixed	0.70	0.588662	0.596810	0.767243
Random	0.10	0.138429	0.242728	0.372060
Random	0.30	0.200572	0.318891	0.447853
Random	0.50	0.325732	0.432183	0.570729
Random	0.70	0.546080	0.584678	0.738972

4 PIT and QQ Diagnostics

These probability integral transform and quantile plots are descriptive diagnostics; they are not formal evidence of joint calibration.

5 PatchTST Detailed Results

The second-backbone study uses one backbone seed and one probability-head seed per dataset. ETTh1 and Weather pass the full paired criteria; Electricity preserves NLL and CRPS improvements but exceeds the 1% point-error tolerance.

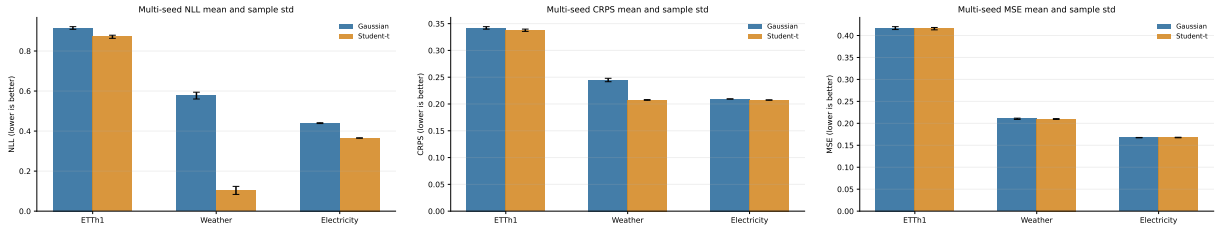


Figure 1: TimePerceiver probability-head seed summaries for NLL, CRPS, and MSE.

Table 4: ETTh1 paired results across additional missing-history conditions. Negative deltas favor fixed Student-t.

Condition	ΔNLL	ΔCRPS	ΔMSE	NLL win	CRPS NI	Point NI
Random 0.1	-0.078025	-0.006113	-0.002231	Yes	Yes	Yes
Random 0.3	-0.041381	-0.003864	-0.001155	Yes	Yes	Yes
Random 0.5	-0.013720	0.000656	0.007100	Yes	No	No
Block 0.3	-0.061086	-0.007738	-0.004834	Yes	Yes	Yes
Channel 0.3	-0.071504	-0.004849	0.001022	Yes	Yes	Yes
Mixed 0.3	-0.070665	-0.003286	0.001169	Yes	Yes	Yes

6 Adapter Ablation

The static adapter improves on the official reference in the evaluated ETTh1 condition, while global-statistics FiLM does not improve over matched static capacity.

Table 5: Static and missingness-aware adapter comparison on ETTh1 random-0.3.

Variant	Trainable	Overhead	Best epoch	Val. MSE	Test MSE	Test MAE
Official	0	0.00%	NA	NA	0.425394	0.450336
Static adapter	33,600	2.96%	1	0.678666	0.361377	0.390458
Missingness-aware FiLM adapter	39,136	3.44%	5	0.679754	0.363156	0.391077

7 Strict Conformal Negative Result

The additive correction uses a temporally separated calibration set and cannot shrink intervals. It widens the already near-nominal native intervals and does not improve absolute coverage error under this protocol.

8 Protocol and Reproducibility

The chronological splits, train-only normalization statistics, fixed validation masks, fixed test masks, metric implementations, evaluation seeds, configuration files, and checkpoint hashes are retained in the project evidence registry. Input length is 384, prediction length is 96, and CRPS uses 64 Monte Carlo samples with fixed evaluation seed 52025. Test masks are never used during head training, and test labels are never used for model selection.

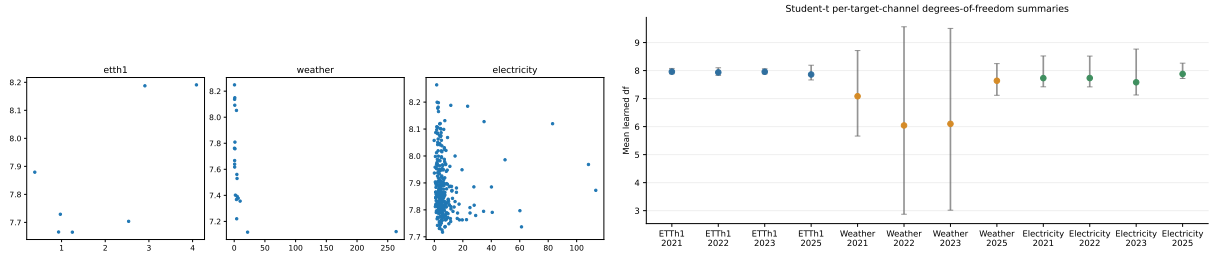


Figure 2: Descriptive channel-tail diagnostics and distribution-head comparison.

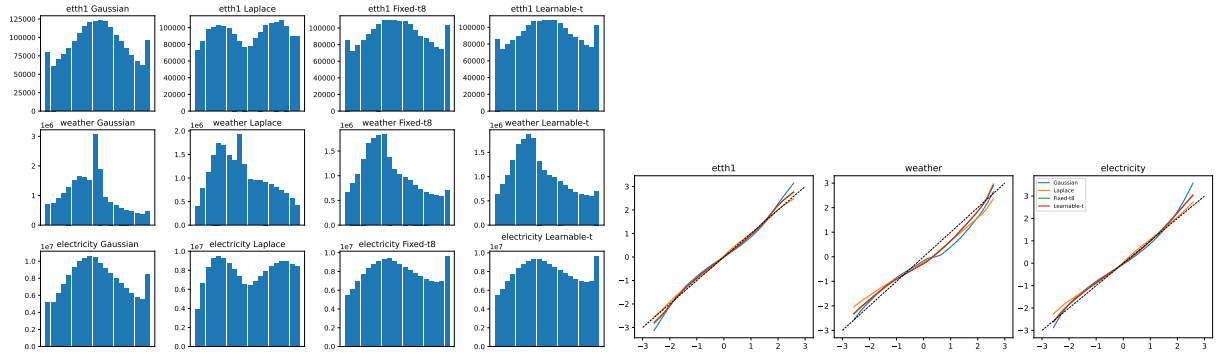


Figure 3: PIT histograms and quantile diagnostics for the distribution ablation.

For traceability, the project stages map to the evidence as follows: the fixed missing-history assets and deterministic evaluation precede the adapter study; probabilistic-head validation and multi-seed fixed-backbone comparisons establish the heavy-tail signal; the distribution ablation refines the final fixed-df specification; the conformal study records a negative result; and PatchTST provides the second-backbone assessment. Historical probability-head runs that did not actually apply the input mask were marked invalid and excluded without reporting their metrics.

9 Source and Mask Integrity

Every formal result is linked to a unique experiment record, resolved configuration, checkpoint hash, fixed-mask hash, and metric-protocol hash. The venue adaptation is a document-only transformation of the frozen evidence: it performs no training, inference, metric recomputation, mask generation, or checkpoint modification. Exact repository URLs and anonymous-review release details remain an author action.

References

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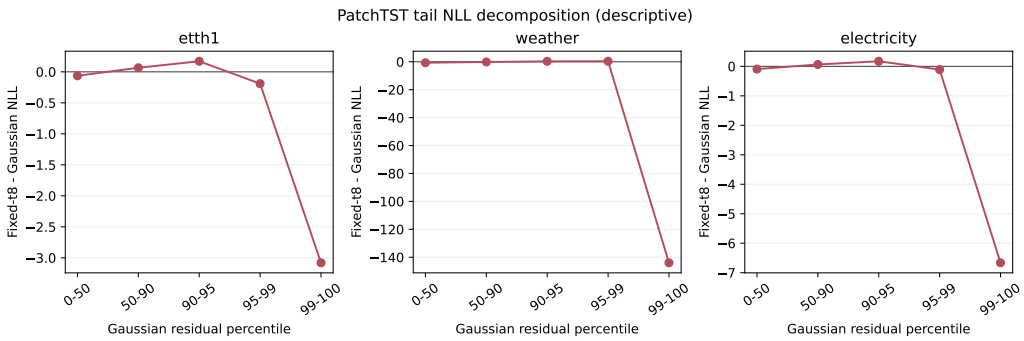


Figure 4: Descriptive PatchTST tail-bin NLL decomposition.

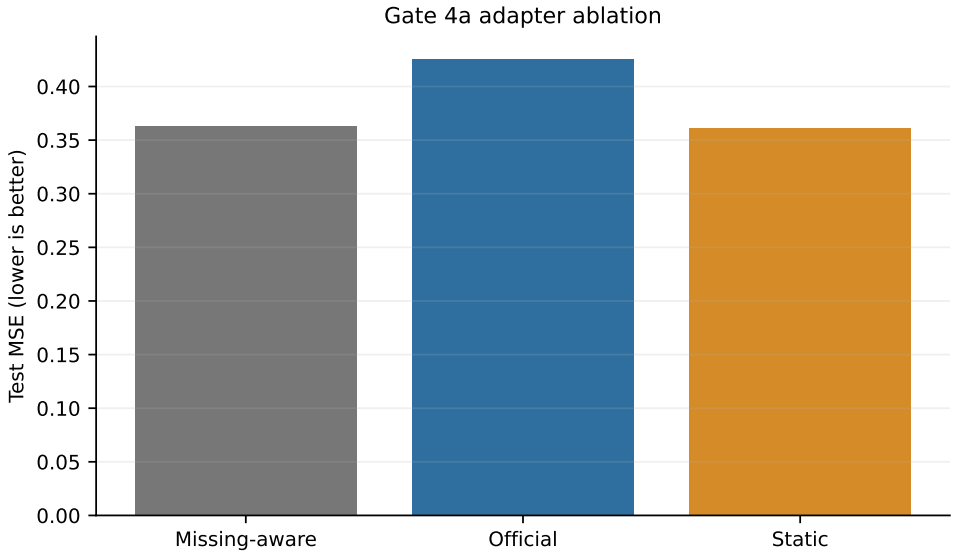


Figure 5: Adapter MSE comparison.

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Table 6: Formal additive conformal results.

Method	Nominal	PICP	MPIW	Coverage error	Interval score
Mask-severity-conditioned conformal	0.90	0.927136	2.262613	0.027136	2.989775
Marginal conformal	0.90	0.927636	2.267989	0.027636	2.990528
Native	0.90	0.897581	1.953189	0.002419	2.946898
Mask-severity-conditioned conformal	0.95	0.964262	2.993500	0.014262	3.625242
Marginal conformal	0.95	0.964505	2.998730	0.014505	3.625259
Native	0.95	0.937700	2.424187	0.012300	3.596873

Table 7: Conformal mechanism audit; counterfactual rows are diagnostic only.

Head	Nominal	Cal. PICP	Test PICP	Shift	$\hat{q}$	Δ coverage error
Gaussian	0.90	0.869114	0.904533	0.035418	0.126842	0.023728
Gaussian	0.95	0.912017	0.937263	0.025246	0.279881	0.001974
Student-t	0.90	0.861667	0.897581	0.035913	0.157400	0.025216
Student-t	0.95	0.913178	0.937700	0.024522	0.287271	0.002205

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Table 8: All retained probability heads and interval procedures.

Head	Method	Nominal	PICP	MPIW	Coverage error	Interval score
Gaussian	Mask-severity-conditioned conformal	0.90	0.927797	2.289391	0.027797	2.979145
Gaussian	Marginal conformal	0.90	0.928261	2.294196	0.028261	2.979930
Gaussian	Native	0.90	0.904533	2.040512	0.004533	2.936425
Gaussian	Mask-severity-conditioned conformal	0.95	0.964530	2.986506	0.014530	3.598434
Gaussian	Marginal conformal	0.95	0.964711	2.991181	0.014711	3.598406
Gaussian	Native	0.95	0.937263	2.431420	0.012737	3.571486
Student-t	Mask-severity-conditioned conformal	0.90	0.927136	2.262613	0.027136	2.989775
Student-t	Marginal conformal	0.90	0.927636	2.267989	0.027636	2.990528
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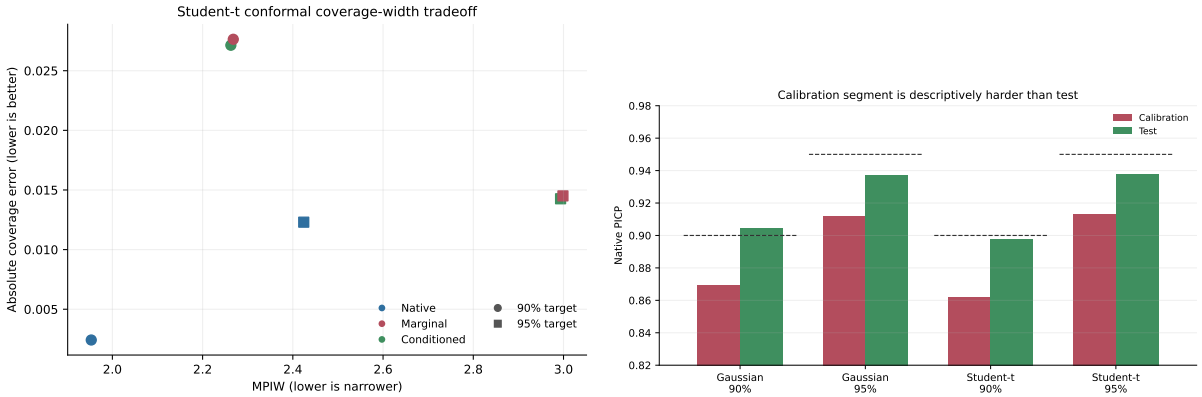


Figure 6: Interval-width tradeoff and calibration-to-test coverage shift.

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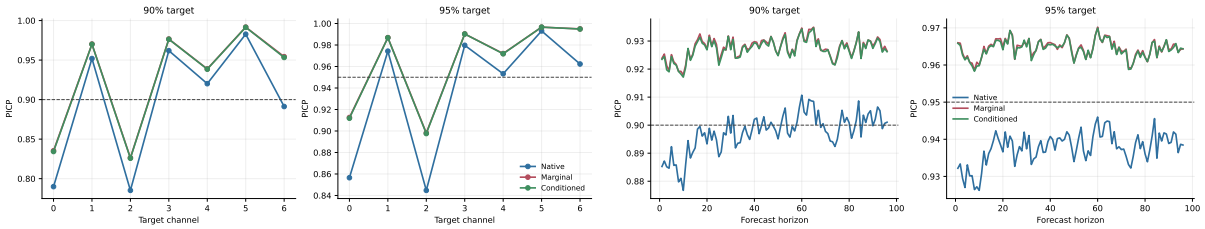


Figure 7: Channel- and horizon-level marginal coverage diagnostics.

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